

Occupational Certificate: Software Developer

SAQA ID

NQF 5

Modelling Processes

251201-005-00-WM02

WORKPLACE SIMULATION – PROJECT REQUIREMENTS

2025



**Creative &
Technology
Universitas**

TABLE OF CONTENTS

REQUIREMENTS	3
PROJECT CONTEXT	3
SYSTEM REQUIREMENTS	3
Pages to Develop	3
Styling Requirements	3
Functional Requirements.....	3

REQUIREMENTS

PROJECT CONTEXT

Edutrack is a growing college network with multiple campuses. They are currently facing operational inefficiencies related to student data management, class scheduling, and academic record handling across their campuses. To streamline these processes, NovaTek Digital has been contracted to design and develop the student portal component of a web-based Student Information Management System (SIMS).

As a junior developer, your role is to design the basic layout and implement core front-end functionality for the SIMS student interface.

SYSTEM REQUIREMENTS

Pages to Develop

- Login / Landing Page
- Registration Page
- Main Dashboard Page
- Uploads Page
- Downloads Page
- Resources Page
- Personal Information Page
- Contact Us Page

Styling Requirements

- Use an external CSS stylesheet
- Maintain consistent styling and layout across all pages
- Include a header banner with the Edutrack logo and navigation links to all pages

Functional Requirements

IndexedDB Integration

- Use IndexedDB to store and retrieve user data

Login / Landing Page

- Users log in using username and password
- Validate input fields (no empty entries)
- Authenticate against credentials stored in IndexedDB
- On failed login → display an alert message
- On successful login → redirect to main dashboard
- Include a link to the Registration Page

Registration Page

- Allow users to create an account and store data in IndexedDB
- Validate all fields (no blank username or password)
- Use regular expressions to enforce password complexity rules

Main Dashboard Page

- Verify user authentication before access
- Display Notice Board (timetable, announcements, etc.)
- Section to display upcoming assessments
- Navigation available in the header

Uploads Page

- Accessible only to logged-in users
- Navigation via banner links
- Section for assessment uploads
- Include placeholder functionality to illustrate upload process

Downloads Page

- Authentication required to access
- Navigation via banner links
- Enable users to view and download uploaded assessments
- Include placeholder download functionality

Resources Page

- Authentication required to access
- Navigation via banner links
- Provide downloadable resource materials (guides, notes, etc.)
- Include placeholder functionality

Personal Information Page

- Authentication required to access
- Navigation via banner links
- Display user data retrieved from IndexedDB
- Allow editing of personal information and password update
- Username must be read-only and cannot be changed

Contact Us Page

- Authentication required to access
- Navigation via banner links
- Display campus contact details
- Include a simplified email/contact form interface